The Regolith Rail Grader

Fusing Loose Dust into Hard Roads, and Ending the Tyranny of Dust

Concept Study RG-CS-033 · Off-World Civilisation Toolkit, Volume II · July 2026
Classification: Speculative engineering, constrained to known physics

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Abstract

Dust is the quiet enemy of every machine that must work on the Moon or Mars. Lunar regolith in particular is sharp, abrasive, electrostatically clinging, and utterly pervasive: it wore out seals and abraded suits within days during the Apollo missions, and it will do worse to a settlement meant to last decades. The single largest generator of that dust is traffic — vehicles grinding across loose ground, throwing it up, packing it into every joint. This study describes a machine that removes the cause rather than fighting the symptom. The RG-1 'Appius' is a self-propelled road-making crawler that levels a strip of regolith and then fuses its surface in place, using concentrated sunlight or microwave heating to sinter the loose grains into a hard, bonded crust — leaving behind a paved road, landing pad, or rail bed made of nothing but the ground it drove over. No binder is imported, no material is hauled; the machine simply cooks the surface it crosses. We describe the sintering principle, the machine, the dust problem it solves, and the honest limits of paving a world one strip at a time.

1. Introduction: The Enemy Is the Ground Itself

Every concept in this catalogue that moves across a planetary surface — the harvesters, the crawlers, the freighters, the mining machines — shares an adversary that has nothing to do with vacuum or radiation or cold. It is dust. Lunar regolith is composed of grains shattered by billions of years of micrometeorite impact, never rounded by wind or water, so they remain jagged and abrasive; they carry electrostatic charge that makes them cling to every surface; and they are fine enough to work into any seal, bearing, or joint. The Apollo crews found dust to be among the most serious practical problems they faced, abrading suit fabric, jamming mechanisms, coating everything, and resisting every attempt to brush it away.

On a surface where nothing binds the ground, the greatest producer of airborne and disturbed dust is traffic itself. A rover crossing loose regolith churns it, throws it, and — in the absence of an atmosphere to slow the ejecta — flings particles on long ballistic arcs that coat distant equipment. Rocket landings are worse still, blasting the unbound surface into a sandblasting storm. The obvious response is to seal, filter, and clean, and those are necessary; but the more powerful response is to stop making the dust in the first place, by binding the ground where vehicles travel and where rockets land. That is the purpose of the Rail Grader.

2. Concept Description

The RG-1 'Appius' is a slow, heavy, self-propelled crawler that lays road by transforming the ground beneath it rather than covering it. It advances along the intended route in a continuous pass. At its front, a blade or grading tool levels the surface, removing rocks and smoothing the profile; behind that, the prepared strip passes beneath a heating head that raises the surface layer of regolith to the temperature at which its grains soften and bond together; and behind that, the fused strip cools into a hard, coherent crust. What emerges from the back of the machine is a solid paved track where loose dust used to be — made entirely of the local ground, with no cement, no binder, and no imported material of any kind.

The heat can be supplied in either of two ways, and the choice defines the machine's character. Concentrated sunlight, gathered by mirrors or lenses and focused onto the surface, is essentially free energy on an airless world with no weather, and it makes the grader a solar machine that works when the sun is up. Microwave heating, driven by electrical power, works regardless of illumination and couples especially well with lunar regolith, whose grains contain nanophase metallic iron that absorbs microwaves strongly, allowing the material to be heated from within at modest power (Taylor and Meek, 2005). A practical grader might carry both — solar concentration for economy in daylight, microwave for the lunar night and for controlled depth.

3. The Physics of Sintering a Road

Sintering is the bonding of a powder into a solid mass by heating it below its full melting point: as the grains are brought to high temperature, material diffuses across the points where they touch, welding them together into a coherent, load-bearing body while the bulk never becomes fully liquid. It is how ceramics and many metal parts are made on Earth, and it applies directly to regolith, which is a powder of mineral grains and glass. The result is not a concrete — nothing is added — but a fused ceramic crust, strong in compression, hard, and, most importantly, bound in place so that it no longer produces dust.

Lunar regolith is unusually cooperative with microwave sintering. Its grains carry nanophase metallic iron produced by micrometeorite impact, and this iron couples strongly with microwave radiation, so lunar soil heats rapidly and from within when microwaved — a property demonstrated on lunar simulant and genuine samples, and one that makes microwave processing of the lunar surface far more efficient than intuition would suggest (Taylor and Meek, 2005). Solar concentration works on a different principle but reaches the same end, delivering intense heat to the surface from above. Either way the physics is the same as the sinter dome concept elsewhere in this catalogue: the ground is the feedstock, heat is the only input, and the product is bonded ceramic — here laid as a ribbon across the landscape rather than raised as a shell.

4. What the Road Is For

A sintered surface serves several distinct purposes, and their relative value shifts as a settlement grows. The first is dust suppression: a bound surface does not billow when driven on, so the traffic that would otherwise be the chief source of dust becomes clean. The second is trafficability: a hard, level road lets vehicles travel faster, carry more, and consume less energy than they can on loose, sinking regolith, and it protects wheels and drives from the abrasion of churned soil. The third, and perhaps most urgent, is the landing pad. A rocket landing on unbound regolith excavates the surface and hurls debris outward at high speed, damaging nearby structures — a hazard that grows with vehicle size, and one that a fused pad largely eliminates.

Beyond roads and pads, the same machine makes rail beds. A sintered strip is a prepared, stable, load-bearing foundation on which rails or guideways can be laid, and rail is the natural mode for heavy, repetitive freight between fixed points — the mine and the refinery, the refinery and the launcher. The grader therefore builds the substrate for the whole surface transport network, from the first cleared path between two habitats to the prepared bed of a mass driver's track. It is unglamorous infrastructure, and it is the kind that everything else quietly depends upon.

Product	Purpose	Value
Paved road	vehicle routes	less dust, faster, less wear
Landing pad	rocket landings	stops debris blasting
Rail bed	heavy freight lines	stable prepared foundation
Work apron	around habitats	clean, bound working surface
Input	the ground itself	no binder, nothing hauled

Table 1. One machine, one feedstock, the whole surface network.

5. Operations Concept

Phase one is the landing pad, which is where fused ground pays for itself fastest: a single sintered apron beneath the arrival point protects every structure nearby from launch and landing debris, and it is a small, bounded area that a modest machine can complete. Phase two is short haul roads — the few hundred metres between habitat, power plant, and mine that carry the most traffic and generate the most dust — progressively binding the ground where vehicles work. Phase three extends to long routes and prepared rail beds, linking distant sites and laying the foundation for heavy freight, including the long, precisely graded bed a mass driver requires. Throughout, the grader works incrementally and its product is permanent: every strip fused is a strip that never needs fusing again, and the network grows by accretion rather than by campaign.

6. Honest Difficulties

The governing constraint is energy per unit area. Sintering demands substantial heat, and a road is a very large area, so paving any real distance is a slow, power-hungry undertaking — this is a machine that creeps, and a settlement will have far less paved ground than it would like for a long time. The fused crust is a ceramic and therefore strong in compression but brittle in tension, so it may crack under concentrated loads or thermal cycling, and it will need a design that tolerates cracking or a reinforcement strategy of the kind the basalt fibre concept supplies. Achieving uniform depth and quality over uneven, variable ground is difficult, and the extreme thermal swings of the lunar day-night cycle will work on the surface over time. The machine itself must survive doing dusty, hot, abrasive work in vacuum for years, which is precisely the environment that destroys mechanisms. None of this is a barrier of physics; regolith sintering is demonstrated and its microwave coupling is well characterised (Taylor and Meek, 2005). They are the honest costs of paving a world with sunlight, one slow strip at a time.

7. Pathway to Reality

The concept sits on solid, tested ground. The severity of the lunar dust problem is documented from Apollo experience onward and is treated as a first-order engineering challenge for surface operations. Sintering regolith with microwaves and with concentrated solar energy has been investigated experimentally, with the strong microwave coupling of lunar soil established and proposals for melting and sintering the surface into roads and pads following naturally from it (Taylor and Meek, 2005). The credible near-term step is a small pad-making unit — a compact, slow machine that fuses a landing apron and demonstrates crust quality, depth control, and durability under real conditions — with road and rail-bed graders following as power budgets allow. Because it directly enables the mass driver, the freighters, and every surface vehicle in this catalogue, the grader is foundational infrastructure: the machine that must run before the others can run well.

8. Conclusion

Roads are the least celebrated and most consequential of technologies, and the first thing a civilisation builds when it means to stay. On the Moon the case is stronger still, because the loose ground is not merely inconvenient but actively hostile — an abrasive, clinging dust that traffic lifts and scatters over everything a settlement owns. The Regolith Rail Grader answers it by cooking the ground into stone as it passes, converting the enemy into the pavement, using nothing but sunlight or electricity and the dust already underfoot. It is slow and hungry for power, and it will never pave as much as anyone wants. But every strip it fuses is a strip that will never blow away again — and a settlement that has bound the ground beneath its wheels has quietly solved the problem that would otherwise have worn it down.

References

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